Power Converter Analysis and Design

ASSIGNMENT - 2

1. An ideal Buck-Boost converter is adjusted such that $V_{out} = -V_{in}$ (assume C.C.M). What must be the duty cycle? ________.

2. An ideal buck converter with $V_{in} = 10\,\text{V}$ is operated at a constant duty ratio of $D = 0.5$ in open loop. If the load resistance is infinity (no-load condition), the output voltage is ________.

3. For steady state operation in a buck converter, the net area under the inductor voltage waveform over one switching period is ________.

4. In an ideal buck converter. The area under the voltage waveform of an inductor over a time interval represents:

   (A) Charge
   (B) Flux density
   (C) Flux linkage
   (D) Stored energy

5. In an ideal buck converter, if the volt-second area of the inductor voltage during the ON interval (DT_s) is smaller than that during the OFF interval ($(1-D)T_s$), then the inductor current will:

   (A) Remain constant
   (B) Increase
   (C) Decrease
   (D) Oscillate around zero

6. The equivalent circuit of a diode in conduction is shown in Fig. 1, modeled with a threshold voltage $V_T = 0.8\,\text{V}$ and a dynamic resistance $r_d = 20\,\text{m}\Omega$. The current through the diode is depicted in Fig. 2, corresponding to a single-phase half-wave rectified load current at a line frequency of $50\,\text{Hz}$. What is the average conduction loss P_{cond} over one fundamental cycle ________.

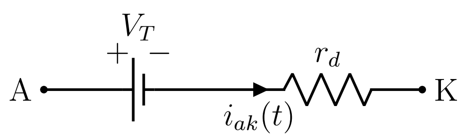

Figure 1: Equivalent circuit of diode while it is conducting.

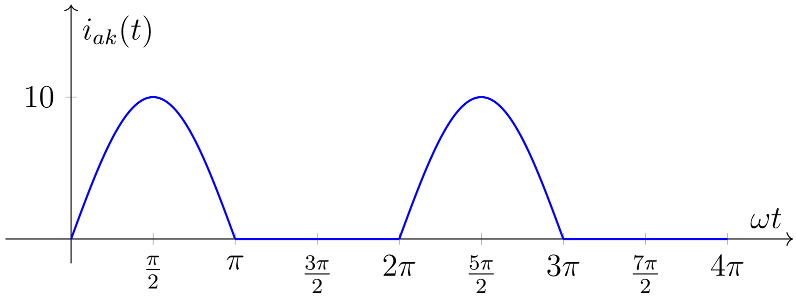

Figure 2: Current waveform through the diode.

7. The turn-off characteristics of a diode are shown in Fig. 3. The diode carries a forward current of $I_O = 20\,\text{A}$ prior to turn-off, with a forward voltage drop of $V_K = 0.9\,\text{V}$. The measured turn-off transition is characterized by $t_o = 50\,\text{ns}$, $t_a = 25\,\text{ns}$, and $t_b = 20\,\text{ns}$. The diode is subjected to a reverse voltage of $V_R = 400\,\text{V}$, and the maximum reverse recovery current is $I_{RM} = 10\,\text{A}$. The switching frequency is $f_s = 50\,\text{kHz}$.

   (A) Calculate the average power loss during the reverse recovery interval $(t_a + t_b)$: ________.

   (B) Calculate the total average turn-off power loss over the entire transition interval $(t_o + t_a + t_b)$: ________.

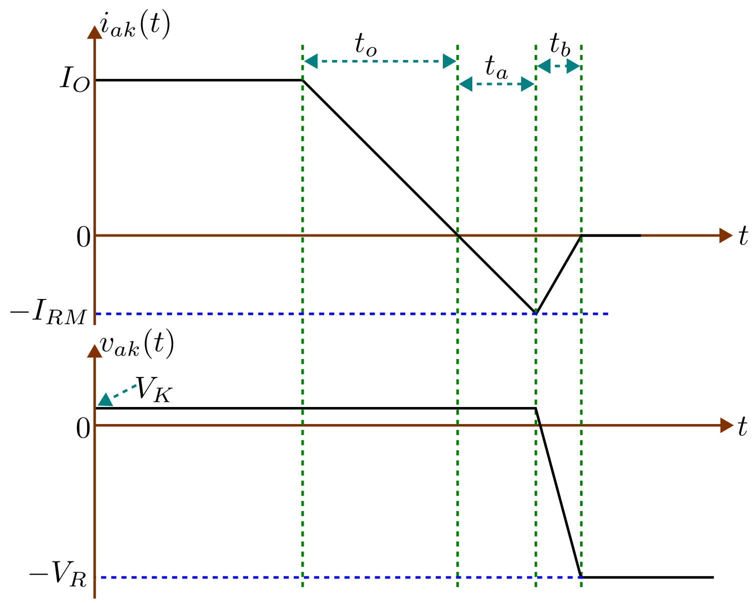

Figure 3: Diode turn off characteristics

8. A power diode exhibits a reverse recovery time of $t_{rr} = 5\,\mu\text{s}$ with a current fall rate of $\dfrac{di}{dt} = 80\,\text{A}/\mu\text{s}$. The softness factor is specified as $SF = \dfrac{t_b}{t_a} = 0.5$, where t_a denotes the time for the reverse current to rise from 0 to I_{RM}, and t_b denotes the time for the current to fall from I_{RM} back to 0, as shown in Fig. 3.

   (A) Calculate the reverse recovery charge, Q_{rr} ________.

   (B) Determine the peak reverse recovery current, I_{RM} ________.

9. A n-channel MOSFET is to be used in a step down (buck) converter circuit. The input dc voltage $V_{in} = 300\,\text{V}$, load current $I_o = 10\,\text{A}$, assume free-wheeling diode is ideal, and the MOSFET is driven by a 30 V (peak-to-peak) square wave (50% duty cycle and zero dc value) in series with $50\,\Omega$. The MOSFET characteristics are $V_{GS(th)} = 4\,\text{V}$, $I_D = 10\,\text{A}$ at $V_{GS} = 7\,\text{V}$, $C_{gs} = 1000\,\text{pF}$, $C_{gd} = 150\,\text{pF}$, and $r_{DS(on)} = 0.5\,\Omega$.

   (A) Sketch and dimension $v_{DS}(t)$ and $i_D(t)$.

   (B) Estimate power dissipation at a switching frequency of 20 kHz.

10. Three MOSFETs, each rated at 2 A, are to be used in parallel to sink a 5 A load current when they are ON. The nominal on-state resistance of the MOSFETs is $2\,\Omega$ at $T_j = 25\,°\text{C}$, but measurements indicate that the actual values for each MOSFET are:

$$r_{\text{DS(on)},1} = 1.8\,\Omega, \quad r_{\text{DS(on)},2} = 2.0\,\Omega, \quad r_{\text{DS(on)},3} = 2.2\,\Omega.$$

    The on-state resistance increases linearly with temperature and is $1.8\times$ larger at $T_j = 125\,°\text{C}$. Determine how much power is dissipated in each MOSFET at a junction temperature of $T_j = 105\,°\text{C}$. Assume a duty cycle of 50%.